



HOISTING FACILITIES

PURPOSE

This document establishes guidelines and recommended procedures for the design of Hoisting Facilities.

SCOPE

This document illustrates the designs of Bridge Crane Girders, Monorails and Jib Cranes/Davit Columns using the latest code references. This guideline does not cover curved monorails which shall be designed according to established principles of mechanics and applicable codes.

APPLICATION

This document deals with the analysis and design of Hoisting Facilities. Hoisting facilities normally encountered in industrial plants are bridge cranes, monorails, and jib cranes.

Bridge cranes and monorails are commonly used in buildings and structures as a means of lifting and moving loads from one place to another for maintenance purposes.

Jib cranes are commonly used to move loads short distances such as in loading or unloading, and for maintenance.

DEFINITIONS

Bridge Crane: A device supported by a pair of support girders to lift large loads and move them from one place to another and commonly used inside buildings.

Monorail: A single beam supporting the hoist unit's bottom flange hanging from connections to its top flange.

Jib Crane:

- Wall bracket jib crane serves as a supplement to regular traveling crane or monorail for smaller loads along the wall or bay. The jib crane rotates up to 200 degrees, serving the area within the radius of the beam's span.
- Free standing jib crane also serves as a supplement to overhead traveling cranes. The free standing jib cranes offer up to 360 degrees rotation without being attached to building structure.



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CMAA 70 CRANE CLASSIFICATIONS

CMAA (Crane Manufacturers Association of America) has classified cranes into categories listed below. The following classifications are taken from AISC Steel Design Guide 7 which it takes directly from CMAA.

Class A (Standby or Infrequent Service)

This service class covers cranes which may be used in installations such as powerhouses, public utilities, turbine rooms, motor rooms and transformer stations where precise handling of equipment at slow speeds with long, idle period between lifts are required. Capacity loads may be handled for initial installation of equipment and for infrequent maintenance.

Class B (Light Service)

This service covers cranes which may be used in repair shops, light assembly operations, service buildings, light warehousing, etc., where service requirements are light and the speed is slow. Loads may vary from no load to occasional full rated loads with two to five lifts per hour, averaging 10 ft per lift.

Class C (Moderate Service)

This service covers cranes that may be used in machine shops or paper mill machine rooms, etc., where service requirements are moderate. In this type of service the crane will handle loads which average 50 percent of the rated capacity with 5 to 10 lifts per hour, averaging 15 ft, not more than 50 percent of the lift at rated capacity.

Class D (Heavy Service)

This service covers cranes which may be used in heavy machine shops, foundries, fabricating plants, steel warehouses, container yards, lumber mills, etc., and standard duty bucket and magnet operations where heavy duty production is required. In this type of service, loads approaching 50 percent of the rated capacity will be handled constantly during the working period. High speeds are desirable for this type of service with 10 to 20 lifts per hour averaging 15 feet, not more than 65 percent of the lifts at rated capacity.

Class E (Severe Service)

This type of service requires a crane capable of handling loads approaching a rated capacity throughout its life. Applications may include custom designed specialty cranes essential to performing the critical work tasks affecting the total production facility.



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These cranes must provide the highest reliability with special attention to ease of maintenance features.

DESIGN CONDITIONS

The vertical and horizontal loads used in the design of Hoisting Facilities will consist of the lifted load plus an impact load. Unless otherwise specified, the impact loads will be taken as follows in accordance with ASCE 7-05 Section 4.10. ASCE 7 is referenced by the International Building Code and is a legal requirement (AISC Steel Design Guide 7). For Situations not covered in ASCE 7-05 refer to AISE (Association of Iron and Steel Engineers) Technical Report No. 13.

Vertical Impact Forces

(ASCE 7 Section 4.10.2)

- Bridge Crane Runway Beams and Connections
 - For cab operated cranes, use 25 percent of the maximum crane wheel loads.
 - For pendant and chain operated cranes, use 10 percent of the maximum crane wheel loads.
- Monorails and Connections
 - For pendant and chain operated cranes, use 25 percent of the lifted load.
- Jib Cranes
 - Use 25 percent of the lifted load.

Lateral and Longitudinal Forces

(ASCE 7 Section 4.10.3)

- Bridge Crane and Monorails
 - Lateral impact of 20 percent of the sum of lifted load, hoist and trolley (exclude bridge weight), applied 1/2 to each runway beam or rails applied perpendicular to runway beam.
 - Longitudinal impact of 10 percent of the maximum crane wheel loads to act parallel to the runway beams.
- Jib Cranes
 - Lateral impact of 25 percent of the lifted load (impact load percentage is defined by this guideline to accommodate the nature of the Jib Crane but increase is not referenced in ASCE 7).



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Load Combinations

The vertical load may be combined with any of the impact loads. However, the vertical and horizontal impact loads will not be considered as acting concurrently. Also, there will be no increase in allowable stress. Vertical and Lateral Impact Loads are considered live loads and should be factored accordingly.

Deflections

The maximum vertical deflections of runway beams, monorails, and jib cranes will not exceed $L/600$ for simply supported beams and $L/300$ for cantilevered beams for CMAA 70 Classes A, B, and C. Impact will not be considered in determining deflections.

The maximum vertical deflection for CMAA 70 Class D cranes is $L/800$ for simply supported beams and $L/400$ for cantilevered beams.

The maximum deflection for CMAA 70 Classes E and F cranes is $L/1000$ for simply supported beams and $L/500$ for cantilevered beams.

Note: *Recommended values for deflection for cantilevered beams are $L/600$ for hoist capacities up to 5 tons and $L/1000$ for larger hoist capacities. This is to reduce the slope the trolley must climb at the end of the cantilever.*

All runway beams loaded laterally with crane lateral loads shall not exceed a maximum deflection greater than $L/400$.

Fatigue

High cycle loaded hoisting facilities will need be designed for fatigue loading. Refer to AISC Steel Construction Manual, 13th Edition Appendix 3 for Fatigue Design.

RUNWAY BEAM DESIGN

The runway beam will normally be designed as a simply supported member with the span equal to the maximum spacing between adjacent supports.

Position the crane to produce the worst loading conditions using Table 3-23 on page 3-226 of the AISC Steel Construction Manual (13th Edition).

Compute the required moment of inertias (I_x and I_y) to satisfy the deflection control criteria.

Calculate the bending moments M_x and M_y including the effects of impact.



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The bending stress will be checked for both single axis and biaxial bending. Make sure to modify member properties for composite members with channel caps.

Refer to *AISC Steel Design Guide 9: Torsional Analysis of Structural Steel Members* on methods to account for torsion in the runway beam. Another method to account for torsion in steel members is provided in *Steel Structures: Design and Behavior* by Salmon & Johnson.

An alternative more crude method of accounting for the eccentric lateral load is provided in *AISC Steel Design Guide 7: Industrial Buildings 2nd ed.* This method considers weak axis bending to be taken by only one flange in flexure. When checking bending stress due to lateral loading, the section modulus of the section under investigation will be taken as 1/2 its elastic value to approximate the effect of torsion for members without channel caps. If there is a channel cap you will need to calculate the section modulus of the top half section. AISC Steel Design Guide 7 Appendix A provides properties for wide flange sections with channel caps.

Note: *The analysis for torsion provided by AISC Steel Design Guide 9 is the preferred method over the method of taking only have the flange in flexure to account for torsion. Consult your project lead in deciding which method to use. The sample designs provided for in the appendix provide a solution for each method.*

It is recommended that brackets or shear connections be used to support runway beams with total reaction at the column of less than 50 kips. Preferably, bearing connections will be used for reactions in excess of 50 kips. Do not use brackets for loads in excess of 50 kips.

MONORAIL DESIGN

The monorail beam will normally be designed as a simply supported member with span equal to maximum spacing of adjacent supports. Frequently, the monorail beam will extend beyond the last point of support. In this case, the beam will be designed as a cantilever from the point of support.

The moment due to vertical loading will be calculated by assuming a point loading at the trolley center.

For the cantilever, the unbraced length of compression flange will be taken as the length of cantilever plus 1/2 the length of the adjacent span.

The lower flange of the monorail beam should be checked for local bending using AISC (13th ed.) Section J10.



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JIB CRANE DESIGN

Jib cranes are either self-supporting or column supported.

Jib crane support columns will be checked for bending and axial load in accordance with AISC Section H1 (13th Ed.).

SPECIAL CONSIDERATIONS

As an integral part of a monorail or bridge crane design, the Structural Engineer should prepare a sketch showing the relative height of items such as runway beams and bridges. This sketch should also include the assumed dimensions for trolleys and hoist.

The Structural Engineer should obtain a copy of the supplier's drawing on all bridge cranes, trolleys, and hoists. The weights and dimensions obtained from these drawings should be compared with the values used in the original design. Any difference in the actual and assumed values should be noted and acted upon accordingly.

RESOURCES

Acco Industries, Inc. Hoist, Trolleys, and Cranes Catalog (Wright). Whiting Crane Handbook.

AISC (American Institute for Steel Construction). Engineering Journal. Fourth Quarter 1982.

Design of Welded Structures. The James F. Lincoln Arc Welding Foundation.

Fisher, James M. and Donald R. Beuttner. Light and Heavy Industrial Buildings. AISC.

Guide For the Design And Construction Of Mill Buildings. AIST/AISE Technical Report No. 13. August 1, 1979.

Reemsnyder, Harold S. and David A. Demo. Fatigue Cracking in Welded Crane Runway Girders, Causes and Repair Procedures. Iron and Steel Engineer. April 1978.



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REFERENCES

ASCE/SEI 7- Minimum Design Loads for Buildings and Other Structures. ASCE (American Society of Civil Engineers). 2005

Carter, Charles j. & Seaburg, Paul A. Steel Design Guide 9: Torsional Analysis of Structural Steel Members. AISC. 1997.

Fisher, James M. Steel Design Guide 7: Industrial Buildings- Roofs to Anchor Rods, 2nd Edition. AISC. 2004.

Steel Construction Manual, 13th Edition. AISC (American Institute for Steel Construction). 2005.

ATTACHMENTS

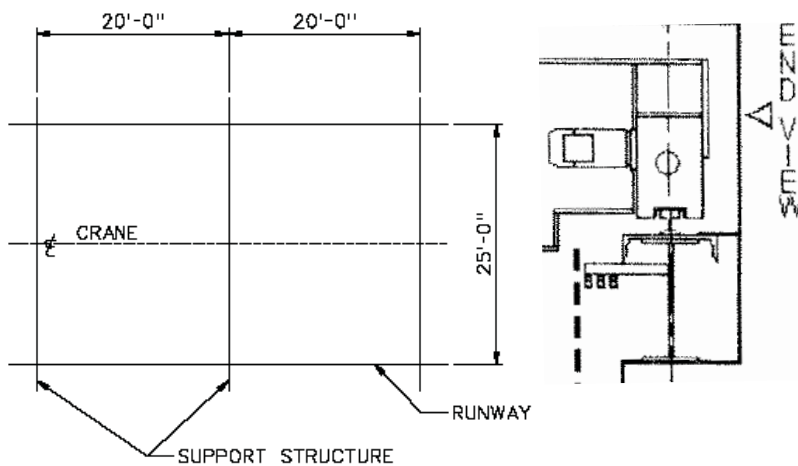
Attachment 01: Sample Design 1: Hoisting Facilities - Bridge Crane Design
Attachment 02:

Attachment 02: Sample Design 2: Hoisting Facilities - Monorail Beam Design

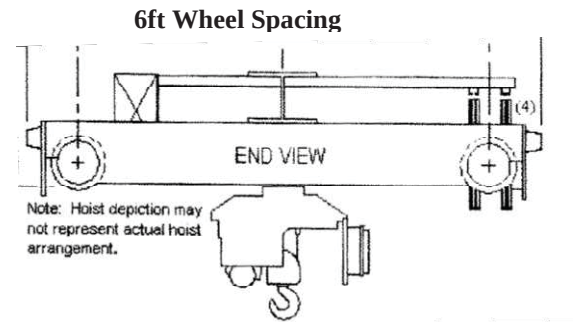


SAMPLE DESIGN 1: HOISTING FACILITIES – BRIDGE CRANE DESIGN

BRIDGE CRANE



PLAN VIEW



END TROLLEY

DESIGN CRITERIA AND REFERENCE

(AISC Sec. 4.10) Use following impact factors for pendant-operated traveling crane support girders and connections.

Vertical Impact - 10% of max. wheel load

Lateral Impact - 20% of lifted load, hoist and trolley
(exclude bridge wt., divided equally between each runway)

Longitudinal Impact - 10% of max. wheel load

DESIGN DATA

(Ref. by crane manufacturer)

Crane capacity = 20.0 K

Bridge WT = 3.26 K

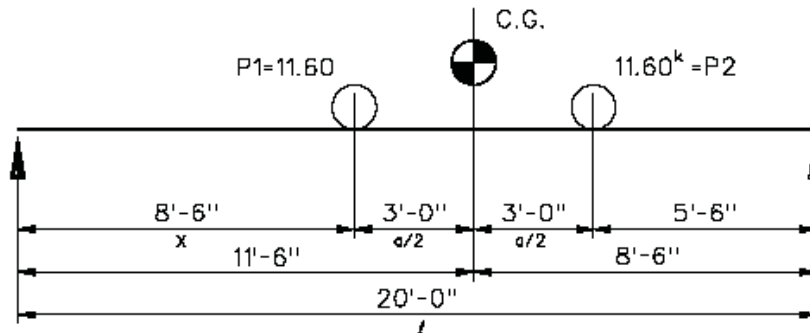
Trolley WT = 0.48 K

Hoist WT = 0.56 K

Wheel load = $(20.0 \text{ K} + 0.48 \text{ K} + 0.56 \text{ K} + 3.26/2 \text{ K})(1/2) = 11.34 \text{ K} < 11.60 \text{ K}$ (max wheel load)



SAMPLE DESIGN 1: HOISTING FACILITIES – BRIDGE CRANE DESIGN



Reference AISC pg. 2-310, 9th EDITION

$$R_1 \text{ max} = P (2 - a/L) = 11.60 (2 - 6/20) = 19.72 \text{ K}$$

$$0.586 (L) = 0.586 (20) = 11.72 > 6.0$$

$$\text{therefore, } x = 1/2 (20 - 6/2) = 8.5'$$

Maximum Load Moment - Vertical

$$M_v = P/2L (L - a/2)^2 = \frac{11.60}{2(20)} (20 - 6/2)^2 = 83.81 \text{ KFT}$$

at $x = 8.6'$,

$$\text{Beam WT} = 0.06(20)^2 \left(\frac{1}{8} \right) = 3.0 \text{ KFT 2}$$

$$= 86.81 \text{ KFT}$$

ADD 10% Impact,

$$+ 8.68 \text{ KFT}$$

$$= 95.50 \text{ KFT (approximately)}$$

Horizontal Load Impact

$$M_h = 1/2 \left(\frac{0.2 \left(\frac{21.04}{2} \right)}{2(20)} \right) \left(20 - \frac{6}{2} \right)^2 = 7.6 \text{ KFT}$$

(each beam)

SAMPLE DESIGN 1: HOISTING FACILITIES – BRIDGE CRANE DESIGN

Allowable Deflections

$$L := 20\text{ft}$$

$$\Delta_{x,\max} := \frac{L}{600} = 0.4\text{ in}$$

$$\Delta_{y,\max} := \frac{L}{400} = 0.6\text{ in}$$

$$P_x := 11.60\text{kips}$$

$$a := \frac{(L - 6\text{ft})}{2} = 7\text{ ft}$$

$$E := 29000\text{ksi}$$

$$I_{x,\text{reqd}} := \frac{P_x \cdot a}{24 \cdot E \cdot \Delta_{x,\max}} \cdot (3 \cdot L^2 - 4 \cdot a^2) = 506.016\text{in}^4$$

For Bridge Crane at Center

$$P_y := 0.20 \frac{(20\text{kips} + 0.48\text{kips} + 0.56\text{kips})}{4} = 1.052\text{kips}$$

$$I_{y,\text{reqd}} := \frac{P_y \cdot a}{24 \cdot E \cdot \Delta_{y,\max}} \cdot (3 \cdot L^2 - 4 \cdot a^2) = 30.594\text{in}^4$$

Top Flange Moment of Inertia Required

Try S15x42.90 w/ C10x15.3

Note: Values taken from AISC 13th ed. Table 1-20 to eliminate time consuming calculations required for plastic section modulus values.

$$W_t := 58.2\text{plf}$$

$$A := 17.1\text{in}^2$$

$$I_x := 615\text{in}^4$$

$$I_y := 81.5\text{in}^4$$

$$S_1 := 65.7\text{in}^3$$

$$S_y := 16.3\text{in}^3$$

$$S_2 := 105\text{in}^3$$

$$r_x := 6.00\text{in}$$

$$r_y := 2.18\text{in}$$

$$Z_x := 87.6\text{in}^3$$

$$Z_y := 25.0\text{in}^3$$

$$t_{w,S} := 0.41\text{in}$$

$$t_{f,S} := 0.62\text{in}$$

$$d_S := 15.0\text{in}$$

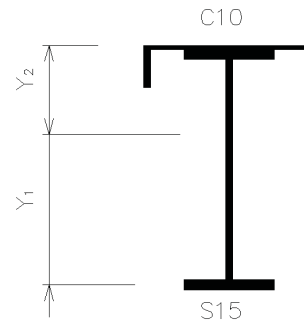
$$d/Af_S := \frac{4.38}{\text{in}}$$

$$t_{w,C} := 0.24\text{in}$$

$$x_{p,C} := 0.224\text{in}$$

$$A_C := 4.48\text{in}^2$$

$$d_C := 10\text{in}$$





SAMPLE DESIGN 1: HOISTING FACILITIES – BRIDGE CRANE DESIGN

$$r_T := 2.94 \text{ in}$$

Values not tabulated in AISC 13th ed. Steel Construction Manual are included in accompanied AISC Manual Companion CD in the AISC 9th ed. ASD Steel Construction Manual.

$$y_1 := 9.37 \text{ in}$$

$$F_y := 50 \text{ ksi}$$

Use A992 Grade Steel for Wide-Flange & A529 Gr. 50 Steel for Channel.

Bending about the Strong Axis

Compaction Check {AISC Table B4.1}

Web

$$d := d_S + t_{w,C} = 15.24 \text{ in}$$

$$y_2 := d - y_1 = 5.87 \text{ in}$$

$$h_C := 2(y_2 - t_{w,C}) = 11.26 \text{ in} \quad \{\text{AISC Spec. Sec. B2.2 (b)}\}$$

$$h_P := h_C = 11.26 \text{ in} \quad \text{Note: } h_P \text{ equal } h_C \text{ because}$$

$$\lambda_{r,w} := 5.7 \cdot \sqrt{\frac{E}{F_y}} = 137.3$$

$$M_y := \min(S_1, S_2) \cdot F_y = 274 \text{ ft} \cdot \text{kips}$$

$$S_{xt} := S_1 = 65.7 \text{ in}^3$$

$$S_{xc} := S_2 = 105 \text{ in}^3$$

$$M_p := \min(Z_x \cdot F_y, 1.6 S_{xc} \cdot F_y) = 365 \text{ ft} \cdot \text{kips}$$

$$\lambda_{p,w} := \frac{\frac{h_C}{h_P} \cdot \sqrt{\frac{E}{F_y}}}{\left(0.54 \frac{M_p}{M_y} - 0.09\right)^2} = 60.678$$

$$\frac{h_C}{t_{w,S}} = 27.4 < \lambda_{p,w} = 60.7 \quad \textbf{**Web is Compact**}$$



SAMPLE DESIGN 1: HOISTING FACILITIES – BRIDGE CRANE DESIGN

Flanges

$$\frac{S_{xt}}{S_{xc}} = 0.626 < 0.70$$

$$\lambda_{p,f} := 0.38 \sqrt{\frac{E}{F_y}} = 9.152$$

$$F_L := F_y \cdot \frac{S_{xt}}{S_{xc}} = 31.29 \text{ ksi} \quad \{\text{AISC Spec. Table B4.1 Footnote (b)}\}$$

$$h := d - 2 \cdot t_{f,S} - t_{w,C} = 13.756 \text{ in}$$

$$k_c := \min \left(\max \left(0.35, \frac{4}{\sqrt{\frac{h}{t_{w,S}}}} \right), 0.76 \right) = 0.691$$

$$\lambda_{r_f} := 0.95 \sqrt{\frac{k_c \cdot E}{F_L}} = 24.05$$

$$t := t_{f,S} + t_{w,C} = 0.862 \text{ in}$$

$$b := \frac{d}{2} = 7.62 \text{ in}$$

$$\frac{b}{t} = 8.84 < \lambda_{r_f} = 24.05 \quad \textbf{**Flange is Compact**}$$

Compression Flange Yielding {AISC F4.1}

$$M_{yc} := S_{xc} \cdot F_y = 437.5 \text{ ft} \cdot \text{kips}$$

$$R_{pc} := \frac{M_p}{M_{yc}} = 0.834$$

$$M_{n_cfy} := R_{pc} \cdot M_{yc} = 365 \text{ ft} \cdot \text{kips}$$



SAMPLE DESIGN 1: HOISTING FACILITIES – BRIDGE CRANE DESIGN

Lateral-Torsional Buckling {AISC F4.2}

$$L_b := 20\text{ft} \quad \text{Unbraced Length}$$

$$L_p := 1.1 \cdot r_T \cdot \sqrt{\frac{E}{F_y}} = 6.49\text{ft}$$

$$h_o := d - \frac{t_{f,S}}{2} - \frac{A_C \cdot x_{p,C} + \frac{d}{d/Af_S} \cdot \left(\frac{t_{f,S}}{2} + t_{w,C} \right)}{A_C + \frac{d}{d/Af_S}} = 14.562\text{in}$$

$$J := 1.75\text{in}^4$$

****Torsional Constant was calculated with RISA Section Program.****

$$L_T := 1.95 r_T \cdot \frac{E}{F_L} \cdot \sqrt{\frac{J}{S_{xc} \cdot h_o}} \cdot \sqrt{1 + \sqrt{1 + 6.67 \cdot \left(\frac{F_L}{E} \cdot \frac{S_{xc} \cdot h_o}{J} \right)^2}} = 28.551\text{ft}$$

$$L_p = 6.49\text{ft} > L_b = 20\text{ft} > L_T = 28.551\text{ft}$$

$$C_b := 1.00 \quad \text{Conservative Case}$$

$$\frac{h_c}{t_{w,S}} = 27.4 < \lambda_{p,w} = 60.7$$

$$M_{n_ltb} := C_b \cdot \left[R_{pc} \cdot M_{yc} - (R_{pc} \cdot M_{yc} - F_L \cdot S_{xc}) \cdot \left(\frac{L_b - L_p}{L_T - L_p} \right) \right] = 309.1\text{ft} \cdot \text{kips} < R_{pc} \cdot M_{yc} = 365\text{ft} \cdot \text{kips}$$

Compression Flange Local Buckling {AISC F4.3}

Flanges are compact therefore the limit state of flange local buckling does not apply



SAMPLE DESIGN 1: HOISTING FACILITIES – BRIDGE CRANE DESIGN

Tension Flange Yielding {AISC F4.4}

$$S_{xt} = 65.7 \text{ in}^3 < S_{xc} = 105 \text{ in}^3$$

$$M_{yt} := F_y \cdot S_{xt} = 273.75 \text{ ft} \cdot \text{kips}$$

$$\frac{h_c}{t_{w,S}} = 27.4 < \lambda_{p,w} = 60.7$$

$$R_{pt} := \frac{M_p}{M_{yt}} = 1.333$$

$$M_{n_tfy} := R_{pt} \cdot M_{yt} = 365 \text{ ft} \cdot \text{kips}$$

Allowable Flexural Strength

$$M_n := \min(M_{n_cfy}, M_{n_ltb}, M_{n_tfy}) = 309.1 \text{ ft} \cdot \text{kips}$$

$$\Omega := 1.67$$

$$M_{a,x} := \frac{M_n}{\Omega} = 185.1 \text{ ft} \cdot \text{kips}$$

$$M_x := 95.5 \text{ ft} \cdot \text{kips}$$

$$\frac{M_x}{M_{a,x}} = 0.516$$

$$\text{if}(M_{a,x} > M_x, \text{"OK"}, \text{"**NG**"}) = \text{"OK"}$$

Biaxial Bending Check

Channel Flexural Strength

From AISC Table 3-8

$$M_{a_Channel} := 62.1 \text{ ft} \cdot \text{kips}$$

With the Channel continuously braced because its weak axis is restrained by the S-section top flange.

S-section Top Flange Flexural Strength

$$Z_{y,S} := 9.08 \text{ in}^3$$

$$S_{y,S} := 5.19 \text{ in}^3$$



SAMPLE DESIGN 1: HOISTING FACILITIES – BRIDGE CRANE DESIGN

$$M_{n_top_flange} := \frac{1}{2} \cdot \min(F_y \cdot Z_{y.S}, 1.6 \cdot F_y \cdot S_{y.S}) = 17.3 \text{ ft} \cdot \text{kips}$$

$$M_{a_top_flange} := \frac{M_{n_top_flange}}{\Omega} = 10.4 \text{ ft} \cdot \text{kips}$$

$$M_{a.y} := M_{a_Channel} + M_{a_top_flange} = 72.5 \text{ ft} \cdot \text{kips}$$

$$M_h := 7.6 \text{ ft} \cdot \text{kips}$$

$$\frac{M_x}{M_{a.x}} + \frac{M_h}{M_{a.y}} = 0.621 \quad \{\text{AISC Eq. H1-1b}\}$$

$$\text{if} \left(\frac{M_x}{M_{a.x}} + \frac{M_h}{M_{a.y}}, "OK", "***NG**" \right) = "OK"$$

Web Sidesway Buckling Check

Use S-Section Properties

$$b_{f.S} := 5.5 \text{ in}$$

$$t_{f.S} = 0.622 \text{ in}$$

$$h = 13.8 \text{ in}$$

$$\frac{\left(\frac{h}{t_{w.S}} \right)}{\frac{L_b}{b_{f.S}}} = 0.767 < 1.7$$

$$1.5 M_{a.y} = 108.7 \text{ ft} \cdot \text{kips} > M_h = 7.6 \text{ ft} \cdot \text{kips}$$

$$C_r := 480000 \text{ ksi}$$

$$R_n := \frac{C_r \cdot t_{w.S}^3 \cdot t_{f.S}}{h^2} \cdot \left[0.4 \left(\frac{\frac{h}{t_{w.S}}}{\frac{L_b}{b_{f.S}}} \right)^3 \right] = 19.77 \text{ kips} \quad \{\text{AISC Eq. J10-7}\}$$



SAMPLE DESIGN 1: HOISTING FACILITIES – BRIDGE CRANE DESIGN

$$\Omega := 1.7$$

$$\frac{R_n}{\Omega} = 11.234 \text{ kips}$$

$$\text{Max_Wheel_Load} := 11.60 \text{ kips}$$

$$\text{if} \left(\text{Max_Wheel_Load} < \frac{R_n}{\Omega}, \text{"OK"}, \text{"**NG**"} \right) = \text{"**NG**"}$$

Try a Section with a thicker web or provide reinforcement. For this example a new section is selected.

Try S15x50 w/ C10x15.3

$$t_{w.S} := 0.550 \text{ in}$$

$$b_{f.S} := 5.64 \text{ in}$$

$$\frac{\left(\frac{h}{t_{w.S}} \right)}{\frac{L_b}{b_{f.S}}} = 0.588 < 1.7$$

$$R_n := \frac{C_r \cdot t_{w.S}^3 \cdot t_{f.S}}{h^2} \cdot \left[0.4 \left(\frac{\frac{h}{t_{w.S}}}{\frac{L_b}{b_{f.S}}} \right)^3 \right] = 21.32 \text{ kips} \quad \{\text{AISC Eq. J10-7}\}$$

$$\frac{R_n}{\Omega} = 12.114 \text{ kips}$$

$$\text{Max_Wheel_Load} := 11.60 \text{ kips}$$

$$\text{if} \left(\text{Max_Wheel_Load} < \frac{R_n}{\Omega}, \text{"OK"}, \text{"**NG**"} \right) = \text{"OK"}$$

USE S15x50 w/ C10x15.3

SAMPLE DESIGN 1: HOISTING FACILITIES – BRIDGE CRANE DESIGN

Girder Tie Back Detail (optional)



Features

- Spherical bearings allow girder end rotation, longitudinal and vertical movement without stressing tie back linkage.
- Single linkage can transmit up to 98 kips side thrust in tension or compression.
- Sizes to fit any girder/column configuration.
- Eliminates diaphragms and the associated maintenance and cracking problems.
- Reference, Gantrex Co., Sales and Warehouse, Pittsburgh (412) 655-1400

SAMPLE DESIGN 2: HOISTING FACILITIES – MONORAIL BEAM DESIGN

DESIGN CRITERIA AND REFERENCE

(ASCE 7 Sec. 4.10)

Vertical Impact - 25% of max. wheel load

Lateral Impact - 20% of lifted load, hoist and trolley
 (Divided equally between each runway)

Longitudinal Impact - 10% of max. wheel load

DESIGN DATA

Monorail Trolley has axle close enough to be
 Considered as a single point

Lifted Load = 4.0 K

Trolley WT + Hoist WT = 0.30 K

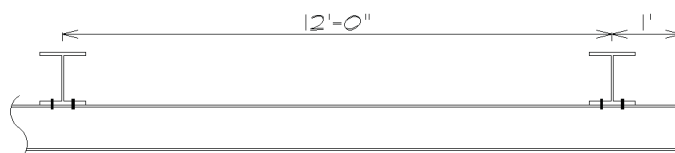
Combined Vertical = 4.3 K

25% Vertical Impact = 1.075 K (ASCE 7 Sec. 4.10.2)

Total Vertical = 5.375 K

20% Lateral Impact = 0.86 K (ASCE 7 Sec. 4.10.3)

10% Longitudinal Impact = 0.43 K (ASCE 7 Sec. 4.10.4)



Assume Interior Span to be designed as a simple beam.

Moments at Cantilever

$L_c := 1\text{ft}$ Cantilever Length

$P_{x,\text{total}} := 5.375\text{kips}$

$P_{y,\text{lat}} := 0.86\text{kips}$

$M_{x,\text{cant}} := P_{x,\text{total}} \cdot L = 5.375\text{ft}\cdot\text{kips}$

$M_{y,\text{cant}} := P_{y,\text{lat}} \cdot L = 0.86\text{ft}\cdot\text{kips}$

$V_{x,\text{cant}} := P_{x,\text{total}} = 5.375\text{kips}$

$V_{y,\text{cant}} := P_{y,\text{lat}} = 0.86\text{kips}$



SAMPLE DESIGN 2: HOISTING FACILITIES – MONORAIL BEAM DESIGN

Allowable Deflection at Cantilever

$$\Delta_{x,\max} := \frac{L}{600} = 0.02 \text{ in}$$

$$\Delta_{y,\max} := \frac{L}{400} = 0.03 \text{ in}$$

$$E := 29000 \text{ ksi} \quad a := 1 \text{ ft} \quad l := 16 \text{ ft}$$

$$I_{x,\text{reqd}} := \frac{P_{x,\text{total}} \cdot a^2}{3 \cdot E \cdot \Delta_{x,\max}} \cdot (1 + a) = 90.7 \text{ in}^4$$

$$I_{y,\text{reqd}} := \frac{P_{y,\text{lat}} \cdot a^2}{3 \cdot E \cdot \Delta_{y,\max}} \cdot (1 + a) = 9.7 \text{ in}^4$$

Moments at Simple Support

$$L := 12 \text{ ft} \quad \text{Simply Supported Span}$$

$$M_{x,\text{simple}} := P_{x,\text{total}} \cdot \frac{L}{4} = 16.125 \text{ ft} \cdot \text{kips}$$

$$M_{y,\text{simple}} := P_{y,\text{lat}} \cdot \frac{L}{4} = 2.58 \text{ ft} \cdot \text{kips}$$

$$V_{x,\text{simple}} := \frac{P_{x,\text{total}}}{2} = 2.688 \text{ kips}$$

$$V_{y,\text{simple}} := \frac{P_{y,\text{lat}}}{2} = 0.43 \text{ kips}$$

Allowable Deflection at Simple Support

$$\Delta_{x,\max} := \frac{L}{600} = 0.24 \text{ in}$$

$$\Delta_{y,\max} := \frac{L}{400} = 0.36 \text{ in}$$

$$E := 29000 \text{ ksi} \quad a := 1 \text{ ft} \quad l := 16 \text{ ft}$$

$$I_{x,\text{reqd}} := \frac{P_{x,\text{total}} \cdot L^3}{48 \cdot E \cdot \Delta_{x,\max}} = 48 \text{ in}^4$$

$$I_{y,\text{reqd}} := \frac{P_{y,\text{lat}} \cdot L^3}{48 \cdot E \cdot \Delta_{y,\max}} = 5.1 \text{ in}^4$$



SAMPLE DESIGN 2: HOISTING FACILITIES – MONORAIL BEAM DESIGN

Try S12x31.8

A992 Steel	$F_y := 50\text{ksi}$				
$A := 9.31\text{in}^2$	$d := 12.0\text{in}$	$t_w := 0.350\text{in}$	$b_f := 5.00\text{in}$	$t_f := 0.544\text{in}$	Gage := 3in
$I_x := 217\text{in}^4$	$S_x := 36.2\text{in}^3$	$r_x := 4.83\text{in}$	$Z_x := 41.8\text{in}^3$		
$I_y := 9.33\text{in}^4$	$S_y := 3.73\text{in}^3$	$r_y := 1.00\text{in}$	$Z_y := 6.44\text{in}^3$		
$r_{ts} := 1.21\text{in}$	$h_o := 11.5\text{in}$	$J := 0.878\text{in}^4$	$C_w := 306\text{in}^6$	$h_o := 11.5\text{in}$	

Torsional Properties {Taken from AISC LRFD Steel Construction Manual 3rd ed.}

$$a := \sqrt{\frac{E \cdot C_w}{G \cdot J}} \quad a := 30.0\text{in} \quad W_{no} := 14.3\text{in}^2 \quad W_{ns} := W_{no} = 14.3\text{in}^2 \quad S_w := 9.74\text{in}^4$$

$$Q_f := 7.24\text{in}^3 \quad Q_w := 20.8\text{in}^3$$

Flexure Strength at Cantilever

Conservatively assume for all cases that $C_b := 1.0$

$$L_b := 5\text{ft}$$

$$M_{a.x.cant} := 120\text{ft} \cdot \text{kips} \quad \{\text{AISC Table 3-7}\}$$

$$\frac{M_{x.cant}}{M_{a.x.cant}} = 0.045$$

$$\Omega := 1.67$$

$$M_{n.y.cant} := \min(F_y \cdot Z_y, 1.6 F_y \cdot S_y) = 24.9\text{ft} \cdot \text{kips}$$

$$M_{a.y.cant} := \frac{M_{n.y.cant}}{\Omega} = 14.89\text{ft} \cdot \text{kips}$$

$$\frac{M_{y.cant}}{M_{a.y.cant}} = 0.058$$

$$\frac{M_{x.cant}}{M_{a.x.cant}} + \frac{M_{y.cant}}{M_{a.y.cant}} = 0.103$$

$$\text{if} \left(\frac{M_{x.cant}}{M_{a.x.cant}} + \frac{M_{y.cant}}{M_{a.y.cant}} < 1.0, \text{"OK"}, \text{"**NG**"} \right) = \text{"OK"} \quad \{\text{AISC Eq. H1-1b}\}$$



SAMPLE DESIGN 2: HOISTING FACILITIES – MONORAIL BEAM DESIGN

Flexure Strength at Simple Support

$$L_b := 12\text{ft}$$

$$M_{a.x.\text{simple}} := 50.1\text{ft}\cdot\text{kips} \quad \{\text{AISC Table 3-7}\}$$

$$\frac{M_{x.\text{simple}}}{M_{a.x.\text{simple}}} = 0.322$$

$$M_{n.y.\text{simple}} := \min(F_y \cdot Z_y, 1.6 F_y \cdot S_y) = 24.9\text{ft}\cdot\text{kips}$$

$$M_{a.y.\text{simple}} := \frac{M_{n.y.\text{cant}}}{\Omega} = 14.89\text{ft}\cdot\text{kips}$$

$$\frac{M_{y.\text{simple}}}{M_{a.y.\text{simple}}} = 0.173$$

$$\frac{M_{x.\text{simple}}}{M_{a.x.\text{simple}}} + \frac{M_{y.\text{simple}}}{M_{a.y.\text{simple}}} = 0.495$$

$$\text{if} \left(\frac{M_{x.\text{simple}}}{M_{a.x.\text{simple}}} + \frac{M_{y.\text{simple}}}{M_{a.y.\text{simple}}} < 1.0, \text{"OK"}, \text{"**NG**"} \right) = \text{"OK"} \quad \{\text{AISC Eq. H1-1b}\}$$

Torsion at Cantilever Section

Limit State of Yielding for Members under Torsion and Combined Stresses
{AISC 13th ed. Sec. H3.3}

$$\Omega := 1.67 \quad \sigma_a := \frac{F_y}{\Omega} = 29.94\text{ksi} \quad \tau_a := \frac{0.6 F_y}{\Omega} = 17.96\text{ksi}$$

$$T := P_{y.\text{lat}} \cdot \frac{d}{2} = 0.43\text{ft}\cdot\text{kips} \quad \text{Applied Torsion}$$

$$L := 5\text{ft}$$

$$\frac{L}{a} = 2$$

From AISC Steel Design Guide 9: Torsional Analysis of Structural Steel Member
 Appendix B, **Case 9**, $\alpha = 1.0$.

$$G := 11200\text{ksi}$$



SAMPLE DESIGN 2: HOISTING FACILITIES – MONORAIL BEAM DESIGN

At connection

$$\left(\frac{GJ}{T} \cdot \frac{1}{a}\right) = 63.525$$

$$\theta_{z_0} := \frac{0}{\left(\frac{GJ}{T} \cdot \frac{1}{a}\right)} = 0 \text{ rad}$$

$$\left(\frac{GJ}{T}\right) = 1.906 \times 10^3 \text{ in}$$

$$\theta'_{z_0} := \frac{0}{\left(\frac{GJ}{T}\right)} = 0 \cdot \frac{1}{\text{in}}$$

$$\left(\frac{GJ}{T} \cdot 5 \cdot a\right) = 285860 \text{ in}^2$$

$$\theta''_{z_0} := \frac{4.8}{\left(\frac{GJ}{T} \cdot 5 \cdot a\right)} = 0.000017 \frac{1}{\text{in}^2}$$

$$\left(\frac{GJ}{T} \cdot a^2\right) = 1715163 \text{ in}^3$$

$$\theta'''_{z_0} := \frac{0}{\left(\frac{GJ}{T} \cdot a^2\right)} = 0 \cdot \frac{1}{\text{in}^3}$$

At free end

$$\theta_{z_1.0} := \frac{1.0}{\left(\frac{GJ}{T} \cdot \frac{1}{a}\right)} = 0.016 \text{ rad}$$

$$\theta'_{z_1.0} := \frac{0.745}{\left(\frac{GJ}{T}\right)} = 0.00039 \frac{1}{\text{in}}$$

$$\theta''_{z_1.0} := \frac{0}{\left(\frac{GJ}{T} \cdot 5 \cdot a\right)} = 0 \cdot \frac{1}{\text{in}^2}$$

$$\theta'''_{z_1.0} := \frac{0.265}{\left(\frac{GJ}{T} \cdot a^2\right)} = 0.00000015 \frac{1}{\text{in}^3}$$

Flange Compression & Tension Stresses

Flange Warping Torsion Stresses

At connection

$$\sigma_{ws_z_0} := E \cdot W_{ns} \cdot \theta''_{z_0} = 6.963 \text{ ksi}$$

At free end

$$\sigma_{ws_z_1} := E \cdot W_{ns} \cdot \theta''_{z_1.0} = 0 \text{ ksi}$$

Flange Vertical Bending Tension and Compression Stresses

At connection

$$\sigma_{b.x_z_0} := \frac{M_{x.cant}}{S_x} = 1.782 \text{ ksi}$$

At free end

$$\sigma_{b.x_z_1.0} := 0 \text{ ksi}$$

Flange Lateral Bending Tension and Compression Stresses

At connection

$$\sigma_{b.y_z_0} := \frac{M_{y.cant}}{S_y} = 2.767 \text{ ksi}$$

At free end

$$\sigma_{b.y_z_1.0} := 0 \text{ ksi}$$

$$\sigma_{total.z_0} := \sigma_{ws_z_0} + \sigma_{b.x_z_0} + \sigma_{b.y_z_0} = 11.512 \text{ ksi}$$

$$< \sigma_a = 29.94 \text{ ksi} \quad \textbf{*OK*}$$

$$\sigma_{total.z_1.0} := \sigma_{ws_z_1} + \sigma_{b.x_z_1.0} + \sigma_{b.y_z_1.0} = 0 \text{ ksi}$$



SAMPLE DESIGN 2: HOISTING FACILITIES – MONORAIL BEAM DESIGN

Web Shear Stresses

Web Saint-Venant Torsion Stresses

At connection

$$\tau_{t_0} := G \cdot t_w \cdot \theta'_{z_0} = 0 \text{ ksi}$$

At free end

$$\tau_{t_1.0} := G \cdot t_w \cdot \theta'_{z_1.0} = 1.532 \text{ ksi}$$

Web Vertical Bending Shear Stresses

At connection

$$\tau_{bw_z_0} := \frac{V_{x.cant} \cdot Q_w}{I_x \cdot t_w} = 1.472 \text{ ksi}$$

At free end

$$\tau_{bw_z_1.0} := 0 \text{ ksi}$$

$$\tau_{w.total_z_0} := \tau_{t_0} + \tau_{bw_z_0} = 1.472 \text{ ksi}$$

$$\tau_{w.total_z_1.0} := \tau_{t_1.0} + \tau_{bw_z_1.0} = 1.532 \text{ ksi}$$

$$< \quad \tau_a = 17.964 \text{ ksi} \quad \textbf{*OK*}$$

Section Rotation

At connection

$$\theta_{z_0} = 0 \text{ degrees}$$

At free end

$$\theta_{z_1.0} = 0.902 \text{ degrees}$$

Flange Shear Stress

Flange Saint-Venant Torsion Stresses

At connection

$$\tau_{t.z_0} := G \cdot t_f \cdot \theta'_{z_0} = 0$$

At free end

$$\tau_{t.z_1.0} := G \cdot t_f \cdot \theta'_{z_1.0} = 2.382 \text{ ksi}$$

Flange Warping Torsion Shear Stresses

At connection

$$\tau_{w.z_0} := \frac{E \cdot S_w \cdot \theta'''_{z_0}}{t_f} = 0 \text{ ksi}$$

At free end

$$\tau_{w.z_1.0} := \frac{E \cdot S_w \cdot \theta'''_{z_1.0}}{t_f} = 0.08 \text{ ksi}$$

Flange Vertical Bending Shear Stresses

At connection

$$\tau_{bf.x.z_0} := \frac{V_{x.cant} \cdot Q_f}{I_x \cdot t_f} = 0.33 \text{ ksi}$$

At free end

$$\tau_{bf.x.z_1.0} := 0 \text{ ksi}$$

SAMPLE DESIGN 2: HOISTING FACILITIES – MONORAIL BEAM DESIGN

Flange Lateral Bending Shear Stresses

At connection

$$\tau_{bf,y,z_0} := \frac{3}{2} \cdot \frac{V_{y.cant}}{2 \cdot t_f \cdot b_f} = 0.237 \text{ ksi}$$

$$\tau_{f.total,z_0} := \tau_{t,z_0} + \tau_{w,z_0} + \tau_{bf,x,z_0} + \tau_{bf,y,z_0} = 0.567 \text{ ksi}$$

At free end

$$\tau_{bf,y,z_1.0} := 0 \text{ ksi}$$

$$\tau_{f.total,z_1.0} := \tau_{t,z_1.0} + \tau_{w,z_1.0} + \tau_{bf,x,z_1.0} + \tau_{bf,y,z_1.0} = 2.462 \text{ ksi} < \tau_a = 17.964 \text{ ksi} \quad \textbf{*OK*}$$

Torsion at Simple Support

From AISC Steel Design Guide 9: Torsional Analysis of Structural Steel Member
Appendix B, **Case 3** $\alpha = 0.5$

$$L := 12 \text{ ft} \cdot \frac{L}{a} = 4.8$$

At Support

$$\left(\frac{GJ}{T} \cdot \frac{1}{L} \right) = 13.234$$

$$\left(\frac{GJ}{T} \right) = 1.906 \times 10^3 \cdot \text{in}$$

$$\left(\frac{GJ}{T} \cdot a \right) = 57172 \text{ in}^2$$

$$\left(\frac{GJ}{T} \cdot a^2 \right) = 1715163 \text{ in}^3$$

$$\theta_{z_0} := \frac{0}{\left(\frac{GJ}{T} \cdot \frac{1}{L} \right)} = 0 \cdot \text{rad}$$

$$\theta'_{z_0} := \frac{0.41}{\left(\frac{GJ}{T} \right)} = 2.151 \times 10^{-4} \cdot \frac{1}{\text{in}}$$

$$\theta''_{z_0} := \frac{0}{\left(\frac{GJ}{T} \cdot a \right)} = 0 \cdot \frac{1}{\text{in}^2}$$

$$\theta'''_{z_0} := \frac{0.1}{\left(\frac{GJ}{T} \cdot a^2 \right)} = 0.00000006 \cdot \frac{1}{\text{in}^3}$$

At free end

$$\theta_{z_0.5} := \frac{0.14}{\left(\frac{GJ}{T} \cdot \frac{1}{L} \right)} = 0.011 \cdot \text{rad}$$

$$\theta'_{z_0.5} := \frac{0}{\left(\frac{GJ}{T} \right)} = 0 \cdot \frac{1}{\text{in}}$$

$$\theta''_{z_0.5} := \frac{-0.5}{\left(\frac{GJ}{T} \cdot a \right)} = -0.00001 \cdot \frac{1}{\text{in}^2}$$

$$\theta'''_{z_0.5} := \frac{0}{\left(\frac{GJ}{T} \cdot a^2 \right)} = 0 \cdot \frac{1}{\text{in}^3}$$



SAMPLE DESIGN 2: HOISTING FACILITIES – MONORAIL BEAM DESIGN

Flange Compression & Tension Stresses

Flange Warping Torsion Stresses

At supports

$$\sigma_{ws_z_0} := E \cdot W_{ns} \cdot \theta''_{z_0} = 0 \text{ ksi}$$

At midspan

$$\sigma_{ws_z_0.5} := E \cdot W_{ns} \cdot \theta''_{z_0.5} = -3.627 \text{ ksi}$$

Flange Vertical Bending Tension and Compression Stresses

At supports

$$\sigma_{b.x_z_0} := 0 \text{ ksi}$$

At midspan

$$\sigma_{b.x_z_0.5} := \frac{M_{x.simple}}{S_x} = 5.345 \text{ ksi}$$

Flange Lateral Bending Tension and Compression Stresses

At supports

$$\sigma_{b.y_z_0} := 0 \text{ ksi}$$

At midspan

$$\sigma_{b.y_z_0.5} := \frac{M_{y.simple}}{S_y} = 8.3 \times 10^3 \text{ psi}$$

$$\sigma_{total.z_0} := \sigma_{ws_z_0} + \sigma_{b.x_z_0} + \sigma_{b.y_z_0} = 0 \text{ ksi}$$

$$\sigma_{total.z_1.0} := \sigma_{ws_z_0.5} + \sigma_{b.x_z_0.5} + \sigma_{b.y_z_0.5} = 10.019 \text{ ksi} < \sigma_a = 29.94 \text{ ksi} \quad \textbf{*OK*}$$

Web Shear Stresses

Web Saint-Venant Torsion Stresses

At supports

$$\tau_{t_0} := G \cdot t_w \cdot \theta'_{z_0} = 0.843 \text{ ksi}$$

At midspan

$$\tau_{t_0.5} := G \cdot t_w \cdot \theta'_{z_0.5} = 0 \text{ ksi}$$

Web Vertical Bending Shear Stresses

At supports

$$\tau_{bw_z_0} := \frac{V_{x.simple} Q_w}{I_x \cdot t_w} = 0.736 \text{ ksi}$$

At midspan

$$\tau_{bw_z_0.5} := 0 \text{ ksi}$$

$$\tau_{w.total_z_0} := \tau_{t_0} + \tau_{bw_z_0} = 1.579 \text{ ksi} < \tau_a = 17.964 \text{ ksi} \quad \textbf{*OK*}$$

$$\tau_{w.total_z_0.5} := \tau_{t_0.5} + \tau_{bw_z_0.5} = 0 \text{ ksi}$$



SAMPLE DESIGN 2: HOISTING FACILITIES – MONORAIL BEAM DESIGN

Section Rotation

At supports

$$\theta_{z_0} = 0 \text{ degrees}$$

At midspan

$$\theta_{z_0.5} = 0.606 \text{ degrees}$$

Flange Shear Stress

Flange Saint-Venant Torsion Stresses

At supports

$$\tau_{t,z_0} := G \cdot t_f \cdot \theta'_{z_0} = 1.311 \text{ ksi}$$

At midspan

$$\tau_{t,z_0.5} := G \cdot t_f \cdot \theta'_{z_0.5} = 0 \text{ ksi}$$

Flange Warping Torsion Shear Stresses

At supports

$$\tau_{w,z_0} := \frac{E \cdot S_w \cdot \theta'''_{z_0}}{t_f} = 0.03 \text{ ksi}$$

At midspan

$$\tau_{w,z_0.5} := \frac{E \cdot S_w \cdot \theta'''_{z_0.5}}{t_f} = 0 \text{ ksi}$$

Flange Vertical Bending Shear Stresses

At supports

$$\tau_{bf,x,z_0} := \frac{V_{x,\text{simple}} \cdot Q_f}{I_x \cdot t_f} = 0.165 \text{ ksi}$$

At midspan

$$\tau_{bf,x,z_0.5} := 0 \text{ ksi}$$

Flange Lateral Bending Shear Stresses

At supports

$$\tau_{bf,y,z_0} := \frac{3}{2} \cdot \frac{V_{y,\text{simple}}}{2 \cdot t_f \cdot b_f} = 0.119 \text{ ksi}$$

At midspan

$$\tau_{bf,y,z_0.5} := 0 \text{ ksi}$$

$$\tau_{f,\text{total},z_0} := \tau_{t,z_0} + \tau_{w,z_0} + \tau_{bf,x,z_0} + \tau_{bf,y,z_0} = 1.624 \text{ ksi}$$

$$< \tau_a = 17.964 \text{ ksi}$$

OK

$$\tau_{f,\text{total},z_1.0} := \tau_{t,z_0.5} + \tau_{w,z_0.5} + \tau_{bf,x,z_0.5} + \tau_{bf,y,z_0.5} = 0 \text{ ksi}$$

SAMPLE DESIGN 2: HOISTING FACILITIES – MONORAIL BEAM DESIGN

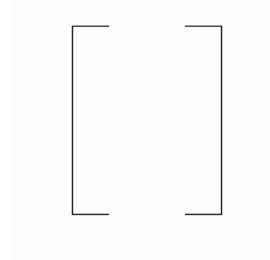
Lower Flange Bending {AISC 13th. ed. Sec J10.1}

$$t_f = 0.544 \text{ in} \quad F_{yf} := F_y = 50 \text{ ksi}$$

$$R_n := 6.25 t_f^2 \cdot F_{yf} = 92.48 \text{ kips}$$

$$\Omega := 1.67$$

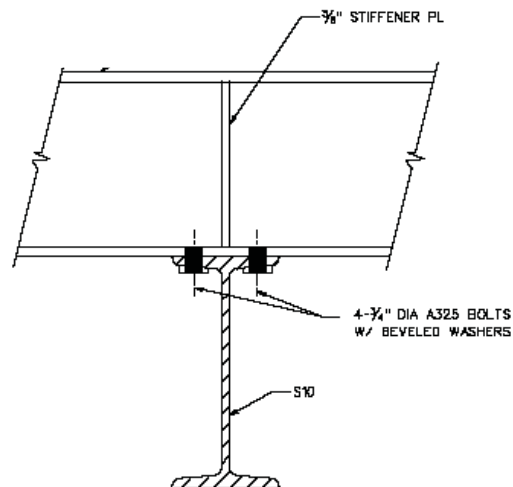
$$\frac{R_n}{\Omega} = 55.377 \text{ kips} > \frac{P_{x,\text{total}}}{2} = 2.7 \text{ kips} \quad \text{for a wheel on each flange}$$



Provide a S12x31.8 Monorail Beam

See figure below for an illustration of to the support girder.

MONORAIL BEAM DESIGN (CONT'D)



Note: The connection has not been designed but simply serves as a guideline.